SMITHIAN FOLD THEORY V3

# Formal Verification Is Not Foundational Derivation

A zero-registered-axiom Smithian Fold audit of OpenAI's ten advances in mathematics and theoretical computer science


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> ***Publication control.*** *This is a new versioned successor. It does not rewrite the preceding paper, its DOI record, historical receipts or evidence. No push, upload, DOI action, release or publication is authorised until Maria Smith explicitly confirms the exact final files and hashes.*
>
> ***Lean verification integration.*** *The current SFT ordered census passed the independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims, 898,902 candidate records and decisions, 11,108 controls, seventeen branches and no reported issue. Lean natively proves the two-class operational root and its unique survivor and constructs proof-bearing acceptance-gate certificates. The remaining claim content is checked as the complete registered artifact graph; that distinction is retained throughout this successor.*

# Abstract

This successor preserves the preceding conclusion-level and admissibility analysis and adds the methodological consequence of SFT's own Lean 4 verification, which passed all 2,777 claims in the current ordered census. Formal acceptance establishes the encoded proposition; SFT admission additionally binds that proposition to its generated grammar, ownership, evidence boundary, candidate decisions, controls and receipts. The new verification does not reclassify any OpenAI comparison verdict or turn external Lean acceptance into automatic SFT theoremhood.

On 1 August 2026 OpenAI announced ten results in mathematics and theoretical computer science and released manuscripts and Lean 4 certificates. This paper freezes the official source at commit `94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6`, resolves the ten advertised advances into twelve main Lean declarations, assigns every declaration to exactly one Smithian Fold Theory branch, and evaluates the artifacts against the already model-admitted SFT theorem-admission law. The upstream manifest reports zero `sorry` uses for every main declaration and reports the same three transitive Lean axioms for every one: `propext`, `Classical.choice`, and `Quot.sound`. That is meaningful evidence of formal completion relative to Lean's foundation. It is not an axiom-free derivation.

The SFT audit is a complete binary product over eight declared theorem-admission dimensions: provenance, registered axioms, proof domain, generality, candidate grammar, controls, independent validation, and admission authority. It enumerates 256 forms per declaration and 3,072 forms in total. For every declaration there is exactly one observed submitted form and exactly one SFT-admissible form; the two differ on all eight dimensions. A separate verifier passes

ten structural checks, including source identity, owner counts, axiom vectors, candidate totals, census hash, and binary disposition.

The closed result is: **12/12 declarations are disproved as submitted SFT theorems, and therefore 10/10 advertised advances are disproved as submitted SFT results. No verdict is OPEN.** The proposition disproved is precisely the proposition that the supplied artifact already has SFT theorem authority. This paper does not claim that Lean derives the negation of the conventional conclusion, nor that a standard Lean axiom is secretly a `sorry`. It shows that kernel checking relative to imported foundations cannot be presented as premise-free, parameter-free, root-traced derivation under SFT.

The comparative-evidence argument is not based on axiom count alone. SFT binds its zero-registered-axiom and zero-free-parameter rule to exhaustive generated alternatives, adverse controls, an immutable admission engine, implementation-distinct reconstruction, preserved failures, and post-seal empirical comparison. Its exact inverse fine-structure result, `503846395469/3676744786`, was sealed before the registered NIST CODATA comparison and lies inside the complete 2022 interval. That does not logically decide an unrelated theorem in combinatorics. It does show why, when the question is foundational derivation with contact to reality, the SFT evidence architecture carries a lower declared assumption burden and a broader falsification surface than a proof certificate alone.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

# Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

# Current status, evidence language and reader map

| Status field | Current position |
|---|---|
| Paper and completion boundary | methodological or comparison scope; no downstream claim ownership transferred. Completion is dated to the current registered census and remains open to lawful versioned extension. |
| Formal status | The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation. |
| Empirical status | Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded. |
| Chronology | Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology. |
| Publication status | Version 0.2.0 is a final local publication candidate awaiting Maria Smith's approval. It is not deposited or published. |
| Ownership | This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law. |
| What is not claimed | Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model. |

## Corpus-wide terminology and evidence key

| Term | Reserved meaning in this paper |
|---|---|
| Theorem | A formally closed proposition at its declared grammar and dependency boundary. |
| Law | An admitted branch relation with its declared carrier, boundary, dependencies and extension rule. |
| Claim | The registered unit judged by the engine and bound to one immutable receipt. |
| Constitution | The rules governing admissible objects, derivations, evidence and publication; not an empirical result. |
| Derivation | Generation and elimination of the declared candidate space to its surviving structure. |
| Prediction | A consequence sealed before the matching target is released under the registered custody protocol. |
| Observation | A source-bound external record; it does not enter candidate generation. |
| Measurement | An instrument-, method-, condition- and uncertainty-bound observed value. |
| Reconstruction | A separately implemented regeneration or an explicitly identified inference of a retained state or history. |
| Exact numerical correspondence | Equality or registered interval relation between exact numerical objects at the declared boundary. |
| Structural correspondence | A post-derivation relation between forms without a claim of exact numerical prediction. |
| Boundary correspondence | Agreement restricted to the named interface, limit or ownership boundary. |
| Compatibility | Non-adverse but non-discriminating evidence. |
| Support | Relevant evidence that is not unique confirmation. |
| Confirmation | Used only when the current registered evidence protocol warrants that classification. |
| Validation | Successful execution of the named formal, computational or empirical test; its kind must be stated. |
| Adverse result | A registered test result that conflicts with the tested claim at its declared boundary. |
| Unresolved result | Required evidence remains unavailable, incomplete, disputed or insufficiently classified. |
| Implementation identity | The hash-bound identity of executable material; implementation success is not empirical confirmation. |
| Formal, empirical and publication status | Three independent classifications; none substitutes for another. |
| Foundational closure | Completion of the registered foundation only; not field-wide closure. |
| Field-wide closure | Completion of the named frozen or registered dated field census. |
| Current-evidence closure | Closure only to the evidence surface presently registered and preserved. |
| Extension openness | New questions may enter a later version without rewriting receipts or history. |

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

## Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

## Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

# 1. The proposition under judgment

The central dispute becomes rigorous only after the proposition is typed. Let `D_i` be one of the twelve supplied Lean declarations, and let

> `S_i` *mean: "The supplied declaration* `D_i`*, with its submitted proof artifact and dependencies, is an admissible theorem of SFT."*

This paper proves `not S_i` for every `i` from one through twelve. The ten OpenAI advances are bundles of those declarations, so the corresponding ten submitted-SFT-result propositions are also false.

This is not wordplay. A theorem is never foundation-free merely because a kernel accepts a term. Kernel acceptance establishes a conditional result: the conclusion follows in the kernel's calculus from the definitions, imported declarations, and axioms on which the term transitively depends. Lean's own reference manual explains that axiom-dependent proof is trusted only to the extent that the axioms are true and mutually consistent, and that `#print axioms` exposes transitive dependence. The same manual identifies `Classical.choice`, `propext`, and `Quot.sound` as standard Lean axioms. Standard status makes them conventional and explicit; it does not make the axiom register empty.

The distinction can be written compactly:

```
Lean-relative question:    Does the conclusion follow in the imported Lean stack?
SFT-admission question:    Was one root-traced, axiom-free, zero-parameter form
                           forced from a complete generated grammar and admitted?
```

The supplied certificates address the first question. This audit decides the second. A negative answer to the second does not erase evidence for the first, but evidence for the first cannot answer the second without changing the definition of SFT theoremhood.

# 2. Frozen source custody

The audit uses only a versioned local capture of official sources. External artifacts are answer-bearing comparison material and have no SFT derivational or admission authority.

| Artifact | Frozen identity | Verification and role |
|---|---|---|
| OpenAI manuscript collection | 249 pages; `sha256:64b900d5fae6fe22f2ae1b8e3b712d20055194a6c81cf343a2455e5898ac7dd6` | Complete released manuscript set; PDF structure passed |
| OpenAI reasoning walkthroughs | 62 pages; `sha256:13b95999f060c0be2142089cfb8b17b75e9231c3c1f3fa0980445ff1b35f0b3b` | Discovery narration; not proof authority |
| Official Lean source archive | Commit `94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6`; `sha256:586a896df5b6a3034bbd2df1eb5ed695100ac7a4ce9eeb486e3e779921667201` | 43 extracted files; archive integrity passed; Apache-2.0 |
| SFT source-custody record | [source_custody_manifest.json](../../experiments/external_sources/mathematics/openai_ten_advances_mathematics_2026-08-01_v1/source_custody_manifest.json) | Records URLs, sizes, hashes, rights, quarantine, and checks |

OpenAI's official release states that an internal system generated the arguments, humans prepared manuscripts with the system, and the model then formalized each argument in Lean. The official repository describes the files as Lean 4 formalizations of the ten results. The precise web sources are listed in the references.

The local custody check did not rerun the complete Lean build. That fact is closed and recorded, not concealed as an unresolved verdict. The upstream `formalization.yaml` itself reports zero sorries and the three axioms for each main result; the fixed source was also inspected at every registered declaration. Rebuilding Lean could add evidence about reproducibility inside Lean, but it could not turn a nonempty published axiom vector into the empty vector required by SFT. Therefore a local build is not a missing premise of the theoremhood disproof established here.

# 3. The pre-existing SFT admission law

This counter-position did not invent a new rule after reading OpenAI's answers. The decisive rule was already registered, generated, sealed, separately reconstructed, and admitted as `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001`. Its exact statement requires the unique admissible route to be root- and source-bound, axiom-free, zero-parameter, completely enumerated, single-survivor, form-closed, adverse-control-tested, independently validated, post-seal with respect to measurements, and engine-admitted before dependency use.

Its model-admission receipt is:

> `sha256:0e21ffcb217271ddfab4000901691761c2b7fd423b93d27f81ab757f0300c58c`

All eight non-empirical gates pass in that receipt; violations are empty. The SFT Constitution independently states the relevant boundaries:

- proof values are the One, exact positive rational parts, generated finite counts, and exact structural carriers;
- semantic numerical zero, negative quantities, irrational proof values, completed infinity, and ungenerated continua are not admitted as derivational objects;
- conventional models and answer-producing priors may frame questions but cannot select an SFT law;
- every assembled law must preserve a complete generated census, minimality, named-shape uniqueness, and a declared closure boundary;
- every admitted claim must have a package and must pass the unchanged admission engine;
- disagreement and adverse evidence are preserved rather than back-fitted away.

In this paper, "zero axioms" therefore means **zero registered imported axioms in the SFT claim package**, and "zero parameters" means **zero free, fitted, or target-selected parameters**. It does not mean that a prose assertion becomes true by calling itself axiom-free. The candidate census, eliminations, controls, seal, validation, and receipt are part of the meaning.

# 4. Complete admissibility enumeration

## 4.1 Declared binary grammar

For each submitted declaration, the audit enumerates the complete Cartesian product of these eight binary dimensions:

| Dimension | Observed submitted coordinate | Required SFT coordinate |
|---|---|---|
| Provenance | imported external stack | SFT root-traced |
| Registered axioms | nonempty registered axioms | empty registered axioms |
| Proof domain | forbidden proof object present | SFT domain only |
| Generality | completed or ungenerated totality | generated finite successor |
| Candidate grammar | no SFT candidate grammar | complete generated census |
| Controls | no SFT adverse-control suite | four registered controls pass |
| Independent validation | Lean agent-review record only | implementation-distinct SFT validation |
| Authority | no SFT engine receipt | model-admitted SFT receipt |

There are `2^8 = 256` audit forms per declaration. Across twelve declarations the exact total is `12 x 256 = 3,072`. This is completeness relative to the declared theorem-admissibility grammar; it is not a claim that 256 forms exhaust every possible future mathematical reformulation of each conventional statement.

The generated census is [candidate_census.json](candidate_census.json), with canonical identity:

```
sha256:6e68afebf8c02b06f7a9b84c5002b2005f71f26349f89e63561f1d13bbffb227
```

## 4.2 Closed proof

**Lemma 1 - Unique required form.** In the complete eight-dimensional binary product, exactly one candidate has the required SFT coordinate on every dimension.

**Proof.** Each dimension has exactly one required coordinate. Their Cartesian product therefore contains exactly one all-required tuple. Direct census count confirms one admissible form per atomic declaration. QED.

**Lemma 2 - Unique observed form.** For each declaration, the fixed upstream artifact instantiates exactly one all-observed tuple.

**Proof.** The source custody fixes one upstream stack, the formalization manifest fixes a nonempty three-axiom vector, source inspection fixes the forbidden-domain carriers, and the submitted package contains no SFT candidate census, SFT adverse-control suite, implementation-distinct SFT validation, or SFT admission receipt. Direct census count confirms one observed form per atomic declaration. QED.

**Lemma 3 - Observed and required forms are unequal.** For each declaration, the unique observed tuple differs from the unique required tuple on all eight coordinates.

**Proof.** Coordinate-by-coordinate comparison in the generated report returns inequality at provenance, axioms, proof domain, generality, candidate grammar, controls, validation, and authority. In particular, the axiom coordinate alone gives a sufficient contradiction: SFT theoremhood implies an empty registered axiom vector, while the supplied artifact reports a nonempty vector. QED.

**Theorem - Universal submitted-artifact rejection.** None of the twelve supplied declarations is an SFT theorem as submitted.

**Proof.** Assume for contradiction that some submitted declaration is an SFT theorem as submitted. It must then instantiate the unique required tuple by the pre-existing admission law. It in fact instantiates the unique observed tuple by source custody and inspection. Lemma 3 proves these tuples unequal. Contradiction. Thus the submitted-SFT-theoremhood proposition is false for every declaration. QED.

**Corollary - Ten advertised results.** Each advertised result contains at least one rejected atomic declaration and no accepted atomic declaration. Therefore all ten are disproved as submitted SFT results. QED.

The machine report is [admissibility_report.json](admissibility_report.json). A separate verifier records `PASS` for all ten checks in [independent_verification_report.json](independent_verification_report.json). Its verified artifact identities are:

- admissibility report: `sha256:1b054b16acde2d3d0e64dc748daee12ad6e46835ba8b3ef1c5ca364381283eb5`;
- candidate census: `sha256:6e68afebf8c02b06f7a9b84c5002b2005f71f26349f89e63561f1d13bbffb227`;
- one-owner ledger: `sha256:c2913c6e952028a26027168d53450cfaf0aabbc50154f331524b111f6b5f1c1d`.

# 5. Exact branch ownership

External comparison is cross-branch; scientific ownership is not. Each atomic declaration has one categorical owner. Typed dependencies may cross branches without moving the owned proposition.

| Atomic result | Categorical owner | Why that owner | Existing typed SFT anchors |
|---|---|---|---|
| OAI26-MATH-001, sphere packing | Mathematics | Geometry, packing, harmonic support, and asymptotic dimension | `SFT-MATH-GEOM-PACKING-COVERING-TESSELLATION-015`; `SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009`; `SFT-MATH-LIMIT-CONTINUUM-002` |
| OAI26-MATH-002, binary-code bound | Mathematics | Coding and asymptotic-rate structure | `SFT-MATH-COMB-CODING-PACKING-010`; rational enclosure; continuum boundary |
| OAI26-MATH-003, spherical-code hierarchy | Mathematics | Coding plus finite-geometric packing | coding/packing; geometry; continuum boundary |
| OAI26-MATH-004, non-sofic group | Mathematics | Group existence and finite approximation | held-inverse group; permutation action; continuum boundary |
| OAI26-MATH-005, Connes rigidity | Mathematics | Group representation and operator-analysis correspondence | representation/action; bounded operator; finite-support integration; continuum boundary |
| OAI26-COMP-001, permanent lower bound | Classical Computation | Formula and circuit resource lower bound | formula branching/circuit; arbitrary Fold-circuit lower bound; symbolic expressions |
| OAI26-QUANTUM-001, parallel repetition | Quantum Computation | Quantum game repetition and parallel resource law | entanglement; quantum parallel complexity; quantum reduction |
| OAI26-COMP-002, GapCVP hardness | Classical Computation | NP-hardness and approximation are computational resource claims | Math lattice/distance inputs; computation reduction and approximation owners |
| OAI26-MATH-006, Ehrhart inequality | Mathematics | Convex and lattice geometry with volume correspondence | convex hull; discrete lattice/polytope; finite-support integration/convergence |
| OAI26-MATH-007, Ramsey lower bound | Mathematics | Extremal combinatorics and graph coloring | Ramsey forcing; coding/packing; graph coloring; continuum boundary |
| OAI26-MATH-008, compactness counterexample | Mathematics | Extremal graph and finite-family structure | extremal set systems; graph packing; continuum boundary |
| OAI26-MATH-009, two-degenerate graphs | Mathematics | Extremal and coloring structure | extremal set systems; coloring; graph packing; continuum boundary |

The count is exact: **9 Mathematics, 2 Classical Computation, 1 Quantum Computation**. GapCVP consumes Mathematics definitions of lattices and distance, but its hardness proposition remains owned by Classical Computation. Quantum parallel repetition remains in Quantum Computation; it is not reassigned to Physics merely because the conventional proof uses quantum language.

The authoritative allocation is [OPENAI_TEN_ADVANCES_ONE_OWNER_LEDGER_2026-08-02.json](../../audits/OPENAI_TEN_ADVANCES_ONE_OWNER_LEDGER_2026-08-02.json). This audit and this paper are downstream correspondence artifacts. They create **no new model admission**. If a future same-question SFT reconstruction produces a genuinely new theorem, it must enter the identified owner branch as a versioned claim and pass the unchanged admission engine there. No cross-branch prose can bypass that route.

# 6. Twelve atomic verdicts, grouped into ten advances

Every row below is binary. "Disproved" retains the exact theoremhood scope defined in Section 1.

| Advance | Atomic declaration(s) | Decisive submitted incompatibility | Closed verdict |
|---|---|---|---|
| 1 | OAI26-MATH-001 | Three imported axioms; real/irrational continuum; completed dimension limit | DISPROVED AS SUBMITTED SFT RESULT |
| 2 | OAI26-MATH-002 and 003 | Three imported axioms; real asymptotic rates; completed at-infinity carrier | DISPROVED AS SUBMITTED SFT RESULT |
| 3 | OAI26-MATH-004 | Three imported axioms; completed infinite group and unbounded approximation totality | DISPROVED AS SUBMITTED SFT RESULT |
| 4 | OAI26-MATH-005 | Three imported axioms; countable infinity, complex operator algebra, continuum measure | DISPROVED AS SUBMITTED SFT RESULT |
| 5 | OAI26-COMP-001 | Three imported axioms; complex field, semantic zero/negative/division, real-log bound | DISPROVED AS SUBMITTED SFT RESULT |
| 6 | OAI26-QUANTUM-001 | Three imported axioms; real continuum value, complex amplitudes, negative exponential values | DISPROVED AS SUBMITTED SFT RESULT |
| 7 | OAI26-COMP-002 | Three imported axioms; conventional NP-hardness stack, signed integer lattice, Euclidean asymptotics | DISPROVED AS SUBMITTED SFT RESULT |
| 8 | OAI26-MATH-006 | Three imported axioms; ungenerated real set, numerical-zero barycenter, continuum volume | DISPROVED AS SUBMITTED SFT RESULT |
| 9 | OAI26-MATH-007 | Three imported axioms; real logarithms/powers and completed at-infinity comparison | DISPROVED AS SUBMITTED SFT RESULT |
| 10 | OAI26-MATH-008 and 009 | Three imported axioms; real fractional powers and unrestricted eventually-at-infinity asymptotics | DISPROVED AS SUBMITTED SFT RESULT |

## 6.1 High-dimensional sphere packing

OpenAI advertises improved asymptotic upper bounds down to the Cohn-Elkies threshold. The main declaration is `PackingBounds.sharpFullCohnElkiesManuscriptConclusions` in `SpherePacking.lean:55591`.

The declaration is stated through real-continuum carriers, `Real.pi`, transcendental functions, and a completed `atTop` limit. Those objects are outside the SFT derivational domain as submitted. More decisively, the proof has the nonempty upstream axiom vector and no SFT admission receipt. The artifact is therefore **disproved as a submitted SFT theorem**. This does not assert that Lean proves the opposite packing inequality; it asserts that the supplied Lean-relative conclusion has not crossed the SFT theorem boundary.

## 6.2 Binary and spherical codes

This advance resolves into two declarations: `MetricCodes.Johnson.binaryRate_lt_mrrw` at `MetricCodes.lean:20797`, and `MetricCodes.Spherical.HigherHierarchy.strict_hierarchy` at `MetricCodes.lean:114336`.

Both use real-valued asymptotic rates and completed length-at-infinity reasoning. Both report the three standard axioms and neither supplies a generated SFT candidate grammar, controls, or receipt. Each atomic theoremhood proposition is false, so the grouped advance is **disproved as a submitted SFT result**. A lawful SFT reconstruction would belong to Mathematics and would need generated finite code families plus a successor or enclosure certificate for any asymptotic correspondence.

## 6.3 A finitely presented non-sofic group

The declaration is `SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group` at `NonSoficGroup.lean:34430`.

Its conventional group framework quantifies over a completed infinite carrier and unbounded finite-approximation totality. Those are meaningful standard mathematical objects, but they are imported rather than forced within the SFT carrier grammar. Together with the nonempty axiom vector and missing SFT receipt, this yields a closed **disproof as a submitted SFT theorem**. The owner is Mathematics.

## 6.4 Connes rigidity

The declaration is `ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors` at `ConnesRigidity.lean:37348`.

The submitted theorem uses a completed countable family, complex Hilbert/operator-algebra machinery, and continuum measure. None is a native SFT proof scalar or generated support as submitted. The artifact also reports the same three axioms. The result is **disproved as a submitted SFT theorem**. That verdict is compatible with the possibility that the conventional counterexample is correct relative to its usual foundations; the two propositions have different proof types.

## 6.5 Permanent formula lower bounds

The declaration is `PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound` at `Permanent.lean:27551`.

It uses a complex scalar field, fraction-field division, semantic zero and negative operations, and a real logarithmic asymptotic resource bound. These are forbidden proof carriers or untranslated correspondences in SFT. Because complexity and lower-bound ownership is computational, this result belongs to Classical Computation, not Mathematics. Its submitted-SFT-theoremhood claim is **disproved**.

## 6.6 Quantum parallel repetition

The declaration is `QuantumParallelRepetition.distributionUniformExponential` at `QuantumParallelRepetition.lean:70953`.

The supplied formulation uses real-continuum strategy values, conventional complex-amplitude quantum machinery, and negative exponential/logarithmic proof values. SFT's Quantum Computation branch instead requires finite exact support, reversible process semantics, and structurally held phase/complement carriers. The nonempty axiom vector supplies an independent decisive failure. The submitted theoremhood claim is **disproved**, and the categorical owner is Quantum Computation.

## 6.7 Closest-vector hardness

The declaration is `GapCVP.Comparator.gapCVP400IsNPHard` at `GapCVP.lean:130398`.

The proposition uses the conventional NP-hardness framework, signed integer lattice coordinates including semantic zero and negatives, Euclidean distance, and asymptotic approximation. Lattice and distance definitions are Mathematics dependencies; the hardness and approximation claim is owned by Classical Computation. With the three imported axioms and no native candidate/receipt route, the artifact is **disproved as a submitted SFT theorem**.

## 6.8 Ehrhart volume

The declaration is `Ehrhart.Volume.ehrhart_volume_inequality_for_sets` at `EhrhartVolumeInequality.lean:55733`.

The conventional statement ranges over an ungenerated real-continuum set, uses a numerical-zero barycenter and lattice point, and invokes continuum volume/integration. SFT can represent finite convex/lattice structures and finite-support integration, but such a translation must itself be forced and certified. It is not supplied here. The submitted theoremhood claim is **disproved** in Mathematics.

### 6.9 Multicolour triangle Ramsey numbers

The declaration is `ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit` at `MulticolorTriangleRamsey.lean:3042`.

The formal bound uses real logarithms, non-rational powers, and a completed at-infinity comparison. These can frame a correspondence target but cannot enter SFT as imported proof objects. The three axioms and missing SFT evidence route close the result as **disproved as a submitted SFT theorem**. A native reconstruction belongs to Mathematics and would require generated colorings and a lawful successor/enclosure certificate.

### 6.10 Extremal compactness and degeneracy conjectures

Two declarations support this advertised advance:
`CompactnessConjecture.quantitativeCompactnessCounterexample` at `CompactnessAndDegeneracy.lean:9320`, and `TwoDegenerateGraphs.twoDegenerateExtremalCounterexample` at `CompactnessAndDegeneracy.lean:18441`.

The submitted formulations use real fractional powers, numerical-zero or negative proposition forms, and unrestricted all-size or eventually-at-infinity claims. Both inherit the three axioms and neither is an admitted SFT result. Each atomic theoremhood claim is false, so the grouped advance is **disproved as a submitted SFT result**. Both are owned by Mathematics.

## 7. What the upstream Lean evidence does establish

A counter-position is strongest when it does not attack a claim the source never made. The official manifest reports `sorry_count: 0` and the same explicit axiom vector for all twelve main declarations. On its own terms, that is a serious formal artifact:

- a zero-sorry declaration does not contain the universal placeholder `sorryAx` according to the upstream manifest;
- a Lean certificate can expose a large proof term to kernel checking;
- `#print axioms` makes transitive foundational commitments auditable;
- source release at a fixed commit makes the submitted artifact inspectable and reproducible in principle.

The correct criticism is therefore not "there is no proof." It is:

> *There is a formal proof relative to an imported conventional stack, but there is no supplied proof that the result is forced from the SFT root with an empty axiom register, zero free parameters, complete native alternatives, adverse controls, and model admission.*

Calling the three axioms "standard" is accurate. Calling the resulting theorem "axiom-free" would be inaccurate. Calling formalization "foundational derivation" without naming the foundation would collapse two different evidentiary claims.

## 8. Comparative evidentiary weight

### 8.1 Five distinct evidence questions

Evidence should be compared by the question it answers.

| Evidence layer | Question answered | OpenAI supplied artifacts | SFT admission architecture |
|---|---|---|---|
| Formal syntax | Is the statement precisely encoded? | Yes, in Lean declarations | Yes, in registered claim packages |
| Relative proof | Does a proof term check under its stack? | Asserted by released Lean certificates; zero sorries in manifest | Engine and proof certificates check registered finite constructions |
| Foundation burden | What is imported rather than derived? | Mathlib/Comparator and three standard axioms are explicit | Zero registered axioms and zero free parameters required per claim |
| Alternative closure | Were all declared rival forms generated and decided? | No SFT-form candidate census supplied | Complete registered grammar, census, eliminations, unique survivor |
| Reality discrimination | Was a sealed consequence tested without target selection? | Not relevant to pure theorem certificates as released | Required wherever the claim has an observable consequence |

Formal verification is exceptionally valuable at the first two layers. SFT asks for the remaining layers before granting its own theorem or law authority. The comparison is therefore not "machine proof versus no machine proof." It is "relative kernel proof" versus "root-traced admission with explicit assumption, alternative, control, and measurement custody."

## 8.2 Exact alpha as a worked reality anchor

The Physics claim `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001` states that a generated binary/generator-three construction forces

```
alpha^-1 = 503846395469/3676744786
```

before measurement. Its registered grammar contains 1,024 forms and exactly one survivor. The model-admission receipt is:

```
sha256:b7b1663a8ed78da756a2624d0bca690ac2ade089a9396060c96f141a6f2896a2
```

That receipt records no axioms, zero free parameters, complete enumeration, unique forcing, form closure, adverse controls, sealing, implementation-distinct reconstruction, and model admission. The decimal below is presentation only; the proof value remains the exact ratio:

```
SFT exact ratio, displayed: 137.0359991771808553263016716765937639926...
NIST CODATA 2022:           137.035999177 +/- 0.000000021
Exact interval:             [137.035999156, 137.035999198]
Central-value difference:   0.0000000001808553263...
Difference / 1 sigma:       approximately 0.008612
```

The separate post-seal validation claim `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001` converts the published decimal and uncertainty to exact rational endpoints and checks interval membership without fitting. Its model-admission receipt is:

```
sha256:13cf55d69e81157f6a00ec4968703133953f817c337147c7d3f160a86d52210a
```

NIST lists `137.035999177(21)` as the 2022 inverse fine-structure constant. The important evidentiary fact is not merely numerical closeness. It is the registered direction of information flow: the SFT ratio seals first; the measurement opens afterward; an adverse changed row must be rejected; the measurement cannot rewrite the survivor.

This alpha result is not a premise in the disproof of the twelve mathematical theoremhood claims. It is comparative evidence for why SFT's zero-parameter standard has empirical meaning rather than functioning as a purely verbal preference.

## 8.3 Constants, values, and provenance

The Physics roadmap explicitly includes "all known physical constants and values at their correct categorical owner" in its dated programme boundary. Correct ownership is not the same as pretending every value has one provenance. The SFT record distinguishes:

- directly forced exact values;
- observationally derived relations;
- separately registered measured values;
- post-seal validations;
- standing predictions without current measurement; and
- adverse or scope-boundary results.

That separation strengthens rather than weakens the zero-parameter position. A measured value is not renamed a first-principles derivation merely to enlarge a headline. A direct derivation such as the exact inverse fine-structure ratio is presented with its own candidate census and seal. The formal Physics Grand Lock `SFT-PHYS-GRAND-LOCK-TERMINAL-075` binds a pre-lock surface of 347 uniquely owned Physics claims and a 534-node transitive dependency dictionary; its receipt is `sha256:ae18f67371c8e7054430935d6b5e5f3162f24cf9cba073769384bf7ba467d817`. The empirical Grand Lock `SFT-PHYS-VALIDATION-GRAND-LOCK-076` binds 234 pre-lock empirical validations, 147 distinct registered external source identities, and all fourteen detected unfavorable or scope-boundary claims; its receipt is `sha256:93f4497d6f7ef3c477246079f62c21f96a7ae27fd9516fa876e9d413bbab569e`.

These counts are evidence-bound inventory claims, not rhetorical synonyms for infallibility. Their value in the present comparison is methodological: provenance and negative evidence remain visible.

### 8.4 Major mathematical problem families

SFT's Mathematics branch reports 323/323 obligations closed at its frozen dated census, including an explicit famous-boundary reconciliation spanning Riemann-mirror, Collatz, Goldbach, twin-prime, continuum/infinity, prime-orbit, and floored-fluid families. The repository also states the exact boundary carried by each result. A finite Collatz census is not silently relabeled as an unrestricted Collatz proof; a Fold-native complexity equality is not silently relabeled as conventional `P = NP`; and a finite-depth fluid result is not silently relabeled as unrestricted continuum Navier-Stokes regularity.

That discipline is part of the persuasive case. A closed model is not one that deletes type boundaries. It is one that returns a determinate disposition at every registered boundary and names the boundary in the conclusion. For the present OpenAI comparison, every registered theoremhood proposition receives a determinate `DISPROVED` verdict. No open verdict is used to hide a mismatch, and no mismatch is inflated into a different theorem.

## 9. Anticipated objections

### 9.1 "These are standard Lean axioms, so they do not count"

They count as exactly what Lean calls them: standard axioms. Their widespread acceptance may justify conventional trust, but the SFT admission law asks whether the registered axiom vector is empty. Three is not zero. The contradiction is definitional and source-backed, not sociological.

### 9.2 "A kernel-checked theorem is simply true"

A kernel checks derivability in a formal environment. Soundness connects that derivability to truth only relative to the consistency and intended interpretation of the environment. Lean's reference documentation itself makes that dependence explicit. Kernel verification is powerful evidence; it is not premise elimination.

### 9.3 "Different foundations cannot refute one another"

This paper does not infer the conventional negation of a theorem from a language mismatch. It refutes the proposition that the supplied artifact already qualifies as an SFT theorem. That proposition is directly decidable because SFT theoremhood has explicit necessary conditions and the artifact has explicit incompatible coordinates. A same-statement SFT proof or counterexample would be a new owner-branch claim, not an implication smuggled through correspondence.

### 9.4 "A zero-axiom label proves nothing by itself"

Agreed. The SFT position does not rest on the label alone. It requires a root trace, complete declared grammar, one survivor, minimality, named-shape uniqueness, adverse controls, a seal, separate reconstruction, and an immutable model-admission receipt. Empirical claims add preregistered post-seal comparison. Any missing gate produces rejection.

### 9.5 "The alpha comparison is irrelevant to pure mathematics"

It is irrelevant as a logical premise and is not used as one. It is relevant to the narrower comparative claim about evidentiary architecture: SFT connects derivation, parameter custody, and external reality in a way a pure theorem certificate is not designed to do. The paper explicitly keeps those roles separate.

### 9.6 "The verdict leaves the real question open"

No registered verdict in this audit is open. Twelve submitted-SFT-theoremhood propositions are false; ten grouped submitted-SFT-result propositions are false. The conventional truth of the conclusions under Lean's foundation is a different proposition, for which the supplied Lean artifacts are evidence and for which this paper asserts no negation. Exact scoping is completion, not evasion.

# 10. Closed disposition

| Disposition item | Count | Final state |
|---|---|---|
| Frozen advertised advances | 10 | 10 DISPROVED AS SUBMITTED SFT RESULTS |
| Frozen main Lean declarations | 12 | 12 DISPROVED AS SUBMITTED SFT THEOREMS |
| Candidate forms enumerated | 3,072 | 3,072 decided within the declared audit grammar |
| Observed submitted forms | 12 | Exactly one per declaration |
| SFT-admissible forms | 12 | Exactly one required form per declaration; none supplied |
| Upstream sorry count | 0 per main declaration, as asserted by upstream manifest | Preserved as favourable upstream evidence |
| Upstream axiom vector | 3 per main declaration | `propext`, `Classical.choice`, `Quot.sound` |
| Owner allocation | 12 | 9 Mathematics; 2 Classical Computation; 1 Quantum Computation |
| OPEN audit verdicts | 0 | None |
| New SFT model admissions created by this audit | 0 | None; future claims must enter their categorical owner branch |

The conclusion is consequently strong and narrow:

> *OpenAI has released substantial Lean-relative proof artifacts. It has not released zero-registered-axiom, zero-parameter, root-traced SFT derivations. Under the pre-existing SFT admission law, every one of the twelve supplied declarations fails theoremhood as submitted, and every one of the ten advertised bundles fails submitted SFT resulthood.*

For researchers who accept Lean's standard foundations, the formalizations deserve direct technical scrutiny on those terms. For a theory whose scientific claim is that law must be forced without imported axioms or fitted parameters, relative formal verification is insufficient. SFT's position carries greater evidentiary weight **for SFT theoremhood and for the stated zero-foundation/zero-parameter question** because it exposes a smaller declared assumption burden, a complete alternative census, negative controls, immutable admission receipts, and, where reality permits, post-seal empirical discrimination.

# 11. Reproducibility

Run from the repository root:

```
python3 frontier/openai_ten_advances_2026/audit_sft_admissibility.py
python3 frontier/openai_ten_advances_2026/independent_verify_admissibility.py
```

Expected audit summary:

```
PASS
declarations 12
advertised_advances 10
candidates 3072
verdict 10/10 DISPROVED AS SUBMITTED SFT RESULTS
```

Expected independent-verification state: `PASS`, with all ten named checks `true`.

The comparison code and generated artifacts are not part of `sft/engine/`, do not change the immutable admission authority, do not enter `census/claims.json`, and do not grant model authority to this paper.

# Appendix A. Frozen declaration inventory

1. `OAI26-MATH-001` - `SpherePacking.lean:55591` - `PackingBounds.sharpFullCohnElkiesManuscriptConclusions`.
2. `OAI26-MATH-002` - `MetricCodes.lean:20797` - `MetricCodes.Johnson.binaryRate_lt_mrrw`.
3. `OAI26-MATH-003` - `MetricCodes.lean:114336` - `MetricCodes.Spherical.HigherHierarchy.strict_hierarchy`.
4. `OAI26-MATH-004` - `NonSoficGroup.lean:34430` - `SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group`.

5. OAI26-MATH-005 - ConnesRigidity.lean:37348 - ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors.
6. OAI26-COMP-001 - Permanent.lean:27551 - PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound.
7. OAI26-QUANTUM-001 - QuantumParallelRepetition.lean:70953 - QuantumParallelRepetition.distributionUniformExponential.
8. OAI26-COMP-002 - GapCVP.lean:130398 - GapCVP.Comparator.gapCVP400IsNPHard.
9. OAI26-MATH-006 - EhrhartVolumeInequality.lean:55733 - Ehrhart.Volume.ehrhart_volume_inequality_for_sets.
10. OAI26-MATH-007 - MulticolorTriangleRamsey.lean:3042 - ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit.
11. OAI26-MATH-008 - CompactnessAndDegeneracy.lean:9320 - CompactnessConjecture.quantitativeCompactnessCounterexample.
12. OAI26-MATH-009 - CompactnessAndDegeneracy.lean:18441 - TwoDegenerateGraphs.twoDegenerateExtremalCounterexample.

Every row has upstream manifest values `sorry_count: 0` and axioms `[propext, Classical.choice, Quot.sound]`.

# Appendix B. Evidence index

## External source custody

- [Local source-capture README](../../experiments/external_sources/mathematics/openai_ten_advances_mathematics_2026-08-01_v1/README.md)
- [Source-custody manifest](../../experiments/external_sources/mathematics/openai_ten_advances_mathematics_2026-08-01_v1/source_custody_manifest.json)
- Captured upstream `formalization.yaml` and `lakefile.toml` under the fixed extracted tree

## SFT audit evidence

- [Admissibility specification](admissibility_spec.json)
- [Generated admissibility report](admissibility_report.json)
- [Generated candidate census](candidate_census.json)
- [Independent verification report](independent_verification_report.json)
- [One-owner ledger](../../audits/OPENAI_TEN_ADVANCES_ONE_OWNER_LEDGER_2026-08-02.json)

## SFT governing and comparative evidence

- [SFT scientific constitution](../../CONSTITUTION.md)
- `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` and receipt `sha256:0e21ffcb217271ddfab4000901691761c2b7fd423b93d27f81ab757f0300c58c`
- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001` and receipt `sha256:b7b1663a8ed78da756a2624d0bca690ac2ade089a9396060c96f141a6f2896a2`
- `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001` and receipt `sha256:13cf55d69e81157f6a00ec4968703133953f817c337147c7d3f160a86d52210a`
- `SFT-PHYS-GRAND-LOCK-TERMINAL-075` and receipt `sha256:ae18f67371c8e7054430935d6b5e5f3162f24cf9cba073769384bf7ba467d817`

- `SFT-PHYS-VALIDATION-GRAND-LOCK-076` and receipt `sha256:93f4497d6f7ef3c477246079f62c21f96a7ae27fd9516fa876e9d413bbab569e`

# References


1. OpenAI, Ten advances in mathematics and theoretical computer science, 1 August 2026.
2. OpenAI, ten-proofs Lean 4 formalizations, accessed 2 August 2026.
3. OpenAI, fixed source commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6, accessed 2 August 2026.
4. Lean project, Axioms - Lean Language Reference, accessed 2 August 2026.
5. Lean project, Axioms and Computation - Theorem Proving in Lean 4, accessed 2 August 2026.
6. NIST, 2022 CODATA recommended values of the fundamental constants, inverse fine-structure constant `137.035999177(21)`.
7. E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, CODATA recommended values of the fundamental physical constants: 2022, Journal of Physical and Chemical Reference Data 54, 033105 (2025).




# Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 0.1.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

On 1 August 2026 OpenAI announced ten results in mathematics and theoretical computer science and released manuscripts and Lean 4 certificates. This paper freezes the official source at commit `94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6`, resolves the ten advertised advances into twelve main Lean declarations, assigns every declaration to exactly one Smithian Fold Theory branch, and evaluates the artifacts against the already model-admitted SFT theorem-admission law. The upstream manifest reports zero `sorry` uses for every main declaration and reports the same three transitive Lean axioms for every one: `propext`, `Classical.choice`, and `Quot.sound`. That is meaningful evidence of formal completion relative to Lean's foundation. It is not an axiom-free derivation.

The SFT audit is a complete binary product over eight declared theorem-admission dimensions: provenance, registered axioms, proof domain, generality, candidate grammar, controls, independent validation, and admission authority. It enumerates 256 forms per declaration and 3,072 forms in total. For every declaration there is exactly one observed submitted form and exactly one SFT-admissible form; the two differ on all eight dimensions. A separate verifier passes ten structural checks, including source identity, owner counts, axiom vectors, candidate totals, census hash, and binary disposition.

The closed result is: **12/12 declarations are disproved as submitted SFT theorems, and therefore 10/10 advertised advances are disproved as submitted SFT results. No verdict is OPEN.** The proposition disproved is precisely the proposition that the supplied artifact already has SFT theorem authority. This paper does not claim that Lean derives the negation of the conventional conclusion, nor that a standard Lean axiom is secretly a `sorry`. It shows that kernel checking relative to imported foundations cannot be presented as premise-free, parameter-free, root-traced derivation under SFT.

The comparative-evidence argument is not based on axiom count alone. SFT binds its zero-registered-axiom and zero-free-parameter rule to exhaustive generated alternatives, adverse controls, an immutable admission engine, implementation-distinct reconstruction, preserved failures, and post-seal empirical comparison. Its exact inverse fine-structure result, `503846395469/3676744786`, was sealed before the registered NIST CODATA comparison and lies inside the complete 2022 interval. That does not logically decide an unrelated theorem in combinatorics. It does show why, when the question is foundational derivation with contact to reality, the SFT evidence architecture carries a lower declared assumption burden and a broader falsification surface than a proof certificate alone.

# Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

# Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

# Limitations and open frontier

The current result is formally complete only to the named dated grammar, census and artifact graph. Native Lean theoremhood is presently established for the operational root and generic gate implications; the other scientific claims are artifact checked rather than individually restated as Lean propositions. Empirical validation remains claim specific, and adverse, unavailable and unresolved evidence remains open where the current records say so. New discoveries, falsifications, stronger comparisons and native claim-level formalisations may enter only through a later version without rewriting this one.

# Data and code availability

The successor Markdown is `frontier/openai_ten_advances_2026/FORMAL_VERIFICATION_IS_NOT_FOUNDATIONAL_DERIVATION_COUNTERPAPER_V0_2.md` and its scientific predecessor is `frontier/openai_ten_advances_2026/FORMAL_VERIFICATION_IS_NOT_FOUNDATIONAL_DERIVATION_COUNTERPAPER_V0_1.md`. The Lean sources and report are under `generated/lean4_validation/`; the report is `generated/lean4_validation/reports/whole_model_validation.json`. Current claims are indexed by `census/claims.json`; complete candidates, decisions, controls, certificates, observations and receipts remain under `claims/<claim-id>/` and `receipts/`. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

# Successor machine-identity appendix

- Preceding source: `frontier/openai_ten_advances_2026/FORMAL_VERIFICATION_IS_NOT_FOUNDATIONAL_DERIVATION_COUNTERPAPER_V0_1.md`
- Preceding source SHA-256: `sha256:a194f9dbc3832520c60bb88f7114fc2c8fe92e081b3c0c231ed48c4b4e1a95ca`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256: `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256: `sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`
- Execution manifest SHA-256: `sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal: `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`
- Verification-authority seal: `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`
- Publication authorised: `false`

- Remote actions performed: `false`

# Lean 4 independent verification update

## Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

| Verified surface | Result |
|---|---|
| Ordered census claims | 2,777 |
| Proof-bearing accepted claim gates | 2,777 |
| Source-bound claims | 2,777 |
| Candidate records | 898,902 |
| Decision records | 898,902 |
| Passed controls | 11,108 |
| Registered branches | 17 |
| Issues | 0 |

This paper is a methodological or comparison paper and does not acquire ownership of downstream claims through this verification.

## Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

## Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

## What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

### Relation to the external Lean comparison

The new SFT Lean layer does not reverse this paper's methodological distinction. A proof assistant checks the proposition encoded for it. Foundational derivation additionally requires that the encoded proposition, grammar, ownership and evidence boundary are the ones lawfully generated by the model. The SFT verifier therefore binds Lean's proof-bearing gates to the registered census, candidate decisions, controls, certificates, source manifests and receipts. It does not treat acceptance of an unrelated Lean declaration as automatic SFT admission, and it does not alter the conclusion-level verdict artifacts audited by this paper.

## Successor conclusion

Version 0.2.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes methodological or comparative analysis and does not acquire ownership of a scientific branch.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.